

Analysis First-Year Exam, May 2023, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let the continuous function $f : (0, \infty) \rightarrow \mathbb{R}$ satisfy

$$\liminf f(n) < 0 < \limsup f(n).$$

Prove that f has infinitely many zeros.

2. Fix any metric space T with exactly two points. Prove that the metric space X is connected if and only if each continuous function f from X to T is constant.
3. Let $B(X)$ be the set of all bounded real-valued functions on the set X . Prove that the formula

$$d(f, g) = \sup\{|f(x) - g(x)| : x \in X\}$$

defines a metric d on $B(X)$ that makes $B(X)$ into a complete metric space.

4. Let $A_1 \supseteq A_2 \supseteq \dots$ be a decreasing sequence of nonempty subsets of $\mathbb{R}^N$ (with its standard Euclidean metric).

(i) Prove that if each A_n is compact then $\bigcap_{n=1}^{\infty} A_n$ is nonempty.

(ii) Show by example that $\bigcap_{n=1}^{\infty} A_n$ can be empty if each A_n is only closed.

5. Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable, with bounded derivative. Prove the existence of the real limits

$$\lim_{x \rightarrow a+} f(x) \quad \text{and} \quad \lim_{x \rightarrow b-} f(x).$$